

CrewAI Multi-Agent Comparison Report

1. Overview

This project evaluated three approaches for an environmental monitoring workflow:

1. **Sequential CrewAI:** Environmental Data Quality Specialist → Remote Sensing Analyst → Environmental Assessment Report Writer
2. **Hierarchical CrewAI:** A manager agent delegates and reviews the work of the specialist agents.
3. **Single-Agent LangGraph:** A single agent-based workflow developed previously in Week 5 Day 3.

The goal was to compare the approaches in terms of output quality, execution cost, reliability, and complexity.

2. Results

Approach	Quality	Token Usage	Latency	Reliability	Complexity
Sequential CrewAI	14/15	21,326	8.07 s	Successful	Medium
Hierarchical CrewAI	Not scored	40,593	Not available	Failed during manager delegation	High
Single-Agent LangGraph	See Day 3 evaluation	See Day 3 results	See Day 3 results	See Day 3 results	Low–Medium

Sequential CrewAI

The sequential crew successfully completed the full workflow. The Data Quality Specialist inspected the dataset, the Remote Sensing Analyst interpreted the environmental measurements, and the Report Writer converted the findings into a stakeholder-ready report.

The run used **21,326 total tokens across 9 successful requests** and completed in approximately **8.07 seconds**. The final output received a manual score of **14/15**, with strong completeness and stakeholder quality.

Hierarchical CrewAI

The hierarchical version used a manager agent to delegate work to the specialist agents. The execution generated **40,593 tokens across 15 successful requests**, which is approximately **1.9× the token usage of the sequential run**.

However, the hierarchical execution failed during manager-level delegation because the delegation tool call did not satisfy the expected schema. Since no complete final report was produced, the run was not assigned a quality score or final latency measurement. This failure is important because it demonstrates that hierarchical delegation introduces additional coordination complexity and additional failure points.

Single-Agent LangGraph

The single-agent LangGraph approach from Day 3 provides a useful baseline because it uses a more centralized workflow rather than distributing responsibilities among multiple CrewAI agents.

Its exact quality, token usage, and latency values should be taken from the Day 3 evaluation rather than estimated or rerun. The main advantage of the single-agent approach is reduced architectural complexity, while the main advantage of the multi-agent approach is clearer specialization of responsibilities.

3. Cost and Quality Analysis

The sequential CrewAI run required significantly fewer tokens than the hierarchical run:

- Sequential: **21,326 tokens**
- Hierarchical: **40,593 tokens**
- Difference: **19,267 tokens**
- Hierarchical usage: approximately **1.9×** sequential usage

Because the internship LLM endpoint does not provide a reliable public per-token price for this experiment, an exact monetary cost is not reported. Token usage is therefore used as the cost proxy.

The sequential crew achieved strong output quality while using fewer tokens and completing successfully. The hierarchical crew consumed more tokens but did not produce a final result, making it less cost-effective for this particular task.

4. Overall Comparison

Factor	Sequential CrewAI	Hierarchical CrewAI	Single-Agent LangGraph
Specialization	High	High	Lower
Workflow control	Explicit	Manager-controlled	Explicit
Flexibility	Moderate	High	Moderate
Token efficiency	Better in this experiment	Worse in this experiment	Expected to be lower complexity
Reliability in this experiment	High	Low	See Day 3 results
Implementation complexity	Moderate	High	Lower
Best suited for	Predictable multi-step workflows	Dynamic tasks requiring delegation	Simple or moderately complex workflows

5. Conclusion

For this environmental monitoring task, the **sequential CrewAI approach was the most effective multi-agent design**. The task naturally follows a fixed sequence of data-quality inspection, analysis, and report generation, so explicit task dependencies worked well without requiring a manager agent.

The hierarchical approach demonstrated the potential for dynamic delegation, but its additional token usage and delegation failure made it less suitable for this particular workflow. A single-agent LangGraph solution would remain attractive when simplicity, lower coordination overhead, and predictable execution are more important than specialization.

Overall, **multi-agent collaboration was worthwhile for this task, but sequential specialization was preferable to hierarchical delegation**. The results also show that adding more agents or a manager does not automatically improve quality, especially when the underlying workflow is already well-defined.